

FoodFighters - Improving Memory Retention of Food Items through a Mobile Serious Game

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Figure 1: Examples of six customized Food Fighters

ABSTRACT

Food waste in private households of industrialised countries is becoming an increasing concern, as its impact on greenhouse gas emissions is comparable to that of the aviation industry. One particular unnecessary contribution to food waste is a person forgetting which food items they have in their household, which often leads to food items expiring before they are used. This paper introduces FoodFighters a mobile serious game aimed at encouraging young adult players to interact with their food items thereby making them more memorable. FoodFighters works by taking pictures of food items, that then can be customized into so called Food Fighters, which are then used to battle other players/friends fighters. In two preliminary lab studies, we validated that Food Fighters actually improved retention of food items and found that players found the game enjoyable based on a GEQ-questionnaire.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**.

KEYWORDS

Food Waste; Sustainability; Serious Games

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1 INTRODUCTION

In recent years we saw a drastic increased effort in the HCI community to develop solutions contributing to a more sustainable future [6, 10, 27, 43]. Examples spanning a wide variety of areas, ranging from interventions to reduce energy consumption [21, 39] to supporting and gamifying recycling behaviour [15, 29]. Furthermore, sustainable food production and consumption-related behaviour has been explored [3, 4, 16] in an attempt to reduce the food-life-cycles direct influence on greenhouse gas emissions and

biodiversity loss. To quantify the impact: World-wide food production accounts for around 26% of world-wide greenhouse gas emissions equivalents (GHGe) [41]. Given that the world's population is increasing, and with that the corresponding need for sustenance, it will be complex to reduce this impact. However, an area that we can improve is food waste, meaning food that is not consumed but rather discarded, either as part of the supply chain or directly by the end-customer. According to Poore and Nemecek 24% of the aforementioned emissions can be attributed directly to food waste [38]. While the majority of this food wasted related GHGe emissions stem from losses in the supply chain, approximately 38% are created by consumer waste. As a result, world-wide approximately 2.2% of all GHGe emissions are produced by food discarded by consumers. In comparison, food waste GHGe production is higher than the aviation industry's (~2%) [38, 41]. As a result, reducing food waste has also been explicitly mentioned as a sub-goal of the United Nations Sustainable Development Goals (SDGs), 12.3 *"By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses"* [36]. So far, the HCI community have focused very little on this SDG [22].

One particular contributor to food waste highlighted by Ganglbauer et al. was forgetting food items in the household. Their approach of augmenting a fridge with the FridgeCam demonstrated how a reminder can help users to reduce this [16]. In this paper, we follow this work and aim to reduce food waste that is a result of forgotten purchases or excess food acquisition. We present FoodFighters, a serious game [33, 37] that – instead of augmenting the fridge – encourages young adult users to take regular pictures of products in their household and thereby aims to improve the retention of knowledge about their food stock. Our preliminary study indicates that FoodFighters has the potential to increase memory retention and thereby potentially prohibit food waste by encouraging the consumption of forgotten food and prevention of stockpiling.

2 RELATED WORK

Recently, research into Human-Food Interaction has seen an increase across a large range of topics [3, 4] even the play potentials of food and eating have been explored [2, 5, 35]. Unfortunately, only little work focused on preventing and mitigating food waste. Prior work focused for example on mobile applications that help to reduce food waste [12, 17]. Ganglbauer et al. developed a mobile food waste diary, to enable users to reflect on moments of food waste and their rationale behind them [17]. Farr-Wharton et al. [12] compared three different applications with differing angles to mitigate food waste. Their results identified food literacy as a crucial aspect of the success of these apps. Ganglbauer et al. analysed the online food-sharing Facebook community of foodsharing.de and highlighted how these kinds of platforms can facilitate fluid changes between global thinking and local action through transitions from online arrangements to offline actions [19]. Ganglbauer et al. also applied the FridgeCam as a technology probe in order to understand food waste in daily behaviour better [16]. Their findings suggest that food waste is often a result of moments of consumption within the users' practices. They present several design opportunities to support the everyday practices of individuals towards reducing food

waste - including making it easier to see what is available and what foods have to be used. Similarly, Farr-Wharton et al. used colour coding and the FridgeCam to raise participants' awareness of available food items in the fridge thereby preventing potential excess shopping and in-time usage of the food [13]. Their findings demonstrate how a properly augmented fridge aiding users' memory and awareness of their food items can assist in preventing food waste. In our work, we follow a similar line of thought by motivating the user to regularly take pictures of the fridge content to increase awareness of food items and subsequently improve information retention of the food items.

With the BinCam, Thieme et al. presented a social persuasive system to motivate reflection on food waste and recycling behaviour of young adults to create behavioural change towards more sustainable practices [8, 9, 47]. Their results indicate that this increased the users' awareness and reflection on their recycling behaviour. Altarriba et al. extended on this idea and created a smart bin concept that uses the power of pranking and social dynamics between groups of students to create awareness of each others food waste [1]. Similarly to the BinCam, it would also post a picture on the social media account of one of the group members [8, 9, 47]. In this work we also incorporate a social aspect, namely playing a game with others, to increase interest through e.g. competitiveness.

Serious Games, i.e., games with serious intent, for example, to educate about a serious problem or to promote behaviour change and to motivate users [33, 37] have been widely applied to a variety of application cases [25, 34, 51]. In the area of sustainable HCI, we have specifically seen a variety of investigations and approaches using serious games to foster more sustainable recycling behaviour in users [7, 15, 23, 26, 29, 31, 40, 42], with some success. Santti et al. reported an increase of the recycling rate of biowaste from 76% to 97% and for recycled plastic from 25% to 84%. Despite such promising results, comparably little work focused on food waste-related serious games [14, 20, 49]. Verloop as well as Gruter focused on data collection about food waste [20, 49] while Ferreira et al. designed a learning game to educate children about the issues regarding food waste [49]. Sinclear et al. developed a provocative serious game, in the form of a quiz game where food would fall into a bin when the users answered questions regarding food literacy wrongly [44]. The here presented work does not focus on food literacy but rather on helping the user to retain memory about fridge contents and thereby mitigating food waste already at the shopping stage.

3 DESIGN & IMPLEMENTATION

The goal of this work was to find a novel and engaging way to reduce food waste in private households. As the target audience we initially chose young students in the age group of 18-35 as Comber et al. and Ganglbauer et al. found that this group, despite having strong positive attitudes towards sustainability, would not always act appropriately with respect to food waste [9, 18]. Our own initial in-situ semi-structured interviews with seven (5 male, 2 female) young adults (mean age 22) who moved out of their parents' house less than three years ago confirmed these findings. We found that their main course of food waste stemmed from forgetting their leftover ingredients. Based on this background, this work aims to

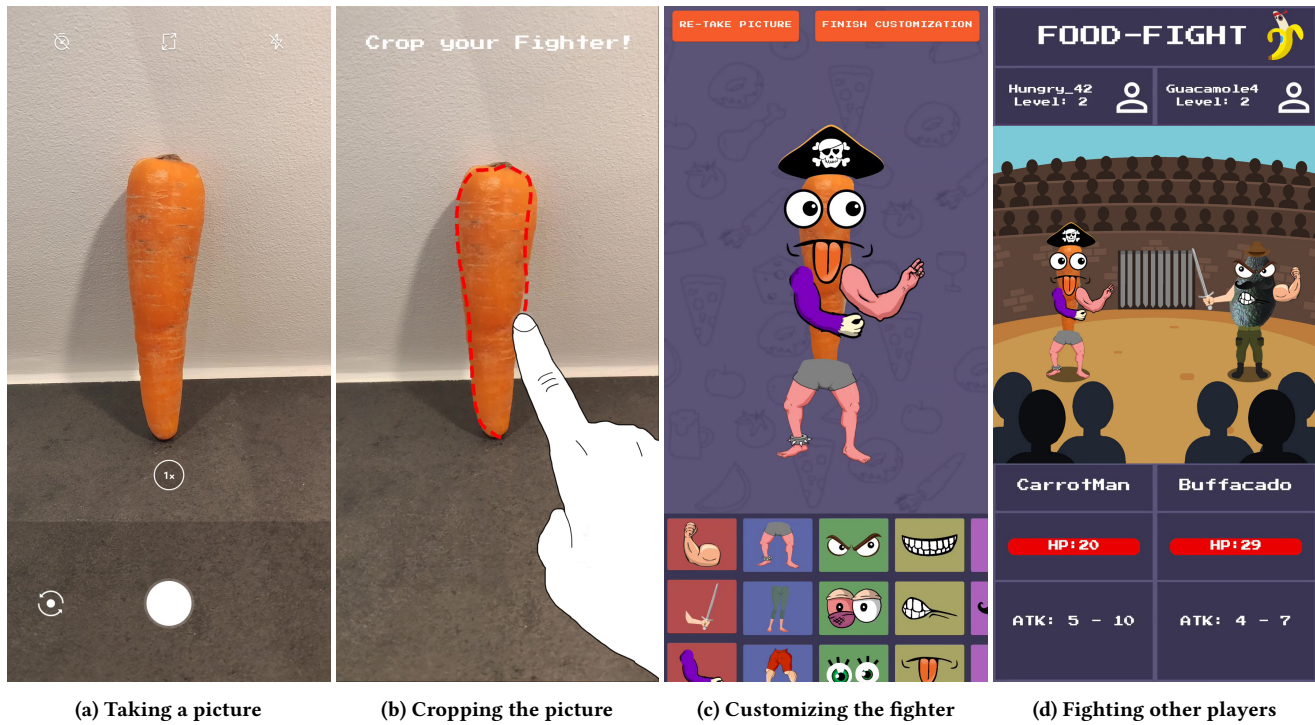


Figure 2: The process of making a Food Fighter, from: 1) taking a picture of a food item, 2) cropping the item out, 3) customizing the fighter, 4) fighting other players' Food Fighters.

help young adults remember food items in their households. To that end, we designed a serious game called FoodFighters, meant to encourage players to interact with their food items more often. Our hypothesis is, that continued interaction with the food after the initial purchase will help users with memory retention (especially for left-overs), and thereby (potentially) prohibit users throwing out or accidentally stockpiling the same items. FoodFighters was implemented as a mobile application for Android phones (to ensure a minimum barrier to usage) and consists of four main parts: a home/menu-screen, customization, gameplay, and social features.

The home screen displays the players' username, level/rank, and previously created Food Fighters, while providing access to Food Fighter creation/customization, fights, and friends.

The customization allows players to personalize their food items into Food Fighters, aimed at creating an emotional attachment, thus making them more memorable [48]. Customization consists of three stages: 1) taking a picture of a food item (see Figure 2a), 2) cropping out the background by drawing an outline around the food item within the frame (Figure 2b), 3) personalizing the food item using six different categories of customization: arms, legs, eyes, mouths, noses/beard, and hats/hair (Figure 2c). Each category has 8-10 unique elements from which players could select a maximum of eight - e.g. allowing for character designs with four arms or two heads (see examples Figure 1). The customization options were presented in a vertical scroll-view. Upon selecting a customization element, players could drag, rotate, mirror, and scale it to their Food Fighter. Once players have named and personalized the physical

characteristics of their Food Fighters, they were then sent to the fight screen to battle other players' Food Fighters (see Figure 2d). Players initially had three energy points to make a new Food Fighter. Once used, it took an hour to recharge an energy point (i.e. if players used all three energies it would take three hours to get them all back). This was done to distribute the playtime into numerous short play-sessions, as opposed to a single long play-session, thus utilizing the concept of spaced learning [45].

The gameplay consisted of each Food Fighter automatically taking turns attacking, thereby damaging their opponent. In the prototype the Food Fighters attack and health point stats were randomized. The fight continues until one of the Food Fighter's health points reaches zero and thereby losses. Each victory accumulates experience points needed to level up the players account. Levels serve as the game's progression unlocking new customization options and allowing the creation of stronger Food Fighters. To incentivize users to continually interact with their food items the Food Fighter permanently dies when loosing a fight. The "dead" is stored in a catalogue showcasing all of the user's previous creations, as well as how well they did in each fight.

FoodFighters utilizes social features such as adding friends whom they can chat, fight, and see their Food Fighters. This encourages users to play alongside friends, enhancing the experience of both [50].

We label FoodFighters as a serious game. While it is in the current version not stated to the user, the serious purpose of FoodFighters is to increase memory retention of the users food supplies, with

the intention to prevent food perishing or stockpiling. Thereby, we argue FoodFighters can be classified as a serious game following Dajouti et al. [11].

4 PRELIMINARY EVALUATIONS

Two preliminary studies evaluated whether the FoodFighter app 1) improves users' retention of food items, and 2) is enjoyable to play.

4.1 Study 1: Retention of food items

The study aimed to test memory retention in a short-term time span of two days to match the time in which food can expire.

4.1.1 Participants. Twenty (14 male, 6 female) students, randomly selected from a campus population, volunteered their time. The 20 participants were equally split into two groups: an experimental group that received the FoodFighter app and a control group who did not.

4.1.2 Procedure. On the first day, participants were deceived that the test concerned their food habits and knowledge. We decided to hide the actual goal of evaluating their ability to remember select food items, in order to prevent them from actively trying to memorize the items. By this, we hope to make it more comparable to a realistic setting, where the users would use the FoodFighter app for the game part and not for memorizing food items. Participants were tasked with simulating two shopping trips. During each shopping trip participants received a grocery list of five items to buy from a basket of 22 food items in-front of them. Participants 'purchased' the items by placing them on a table. After each shopping trip, participants were asked red herring questions about the purchased items: 1) which dish you could make with these ingredients? 2) which item is the most/least climate-friendly? and 3) which item have the longest/shortest expiration date?

On the first shopping trip, participants were asked to purchase: Bacon, ice tea, chopped tomatoes, green pesto, and red onion. Participants in the experimental group then created food fighters from the bacon and red onion. On the second trip, participants purchased: broccoli, cucumber, bell pepper, milk and fish paste. Participants in the experimental group then created food fighters from broccoli and milk.

Two days later, the participants were asked to list all of the 10 items they 'purchased' during the test. The number of items remembered and which items were remembered was noted down by the study facilitator. Additionally, the experimental group provided feedback on the FoodFighter app and whether they had any improvements they would make. All tests were conducted and video recorded in a controlled environment.

4.1.3 Results. A Shapiro-Wilk test showed the number of items remembered was not normally distributed ($p=0.001$), hence we relied on Mann-Whitney U tests to validate significant differences between the two groups.

Out of the ten items, participants with the FoodFighter app tended to ($U=70$, $p=0.1$) to remember more items ($Mdn=9$, $IQR=1.75$), compared to the control group ($Mdn=8$, $IQR=1.75$). The ten items can be further divided into two categories: the four items that the experimental group converted into Food Fighters, and the six items neither groups made into Food Fighters (referred to as non-food

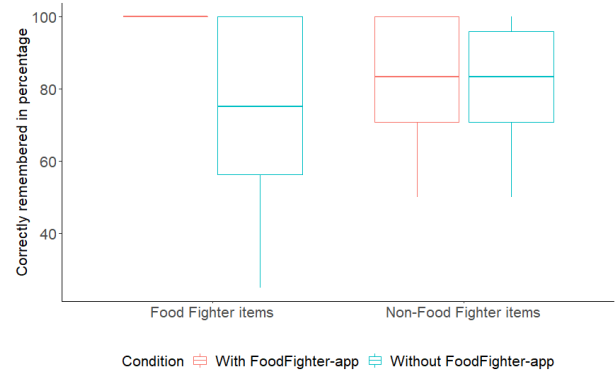


Figure 3: Boxplots displaying the percentage of correctly remembered food items (y-axis). The x-axis shows the two Food Fighter items and non-food fighter items for both the control (blue) and experimental group (red).

fighter items). Participants with the FoodFighter app remembered all four items converted into food fighters while participants without the app remembered significantly ($U=80$, $p=0.005$) less of these items ($Mdn=3$, $IQR=1.75$). The number of non-food fighter items remembered was close to identical ($U=52$, $p=0.9$) for both groups (see Figure 3).

Regarding what the participants thought of the app, half of them expressed concerns with the amount of time each play-session took (especially the fights) and wanted the fights to be more interactive. Nine of the ten participants found the customization of the food items enjoyable. Participants suggested to improve the enjoy-ability of the app, by having new customization items locked behind specific level requirements, and being able to fight with multiple Food Fighters at once.

4.2 Study 2: Enjoyment of the game

4.2.1 Participants. Twenty-three participants (no repeats from study 1) randomly selected from a campus population volunteered their time. Each test was conducted in Danish and translated by the authors.

4.2.2 Procedure. Participants were given the FoodFighter app, and tasked with creating a minimum of two Food Fighters, being able to choose from five different food items (Strawberry jam, Red beans, banana, cucumber, and tomato) and let them fight against simulated other food fighters. Afterwards, they were to complete a GEQ questionnaire using its Core Module [24] on a 5-point Likert scale (0-4). Out of the 7 components within this module, we decided to use the following positive components: *Competence*, *flow*, and *positive affect* and the following negative components: *tension*, and *negative affect*. Additionally, we also asked the participants to describe which aspects they liked as well as which they disliked.

4.2.3 Results. All positive components – competence ($Mdn=2.2$, $IQR=0.6$), flow ($Mdn=2.2$, $IQR=0.8$), and positive affect ($Mdn=2.8$, $IQR=0.7$) – scored above the average of two. While the negative components – Tension ($Mdn=0.3$, $IQR=1.15$), and negative affect ($Mdn=0.5$, $IQR=0.5$) – scored below the average of two (see Figure 4).

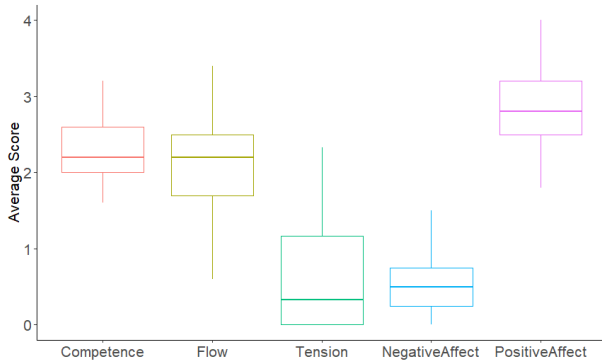


Figure 4: Boxplots displaying the GEQ-test score (min=0, max=4) on the y-axis and the five components: competence, flow, positive affect tension, and negative affect on the x-axis.

Most of the positive feedback was directed at the customization feature, especially due to the number of different items you can put into the Food Fighter. In terms of negative feedback, most of the criticism was mainly directed at the Fight feature due to it happening automatically, and taking too long without any interaction, e.g. P4 stated: *"I felt like I had no agency in the fighting"* Lastly, there seemed to be general confusion in terms of how the health and attack value for the Food Fighters were distributed.

5 DISCUSSION & NEXT STEPS

These evaluations aimed to validate the general concept of FoodFighters. Initial results indicated that the concept will support knowledge retention as participants remembered a significant higher number of food items when using the app. The improved retention could in turn help prevent food waste as more items may get properly consumed instead of forgotten as it was the case in Ganglbauer et al.'s work [17]. To confirm lasting effects on food consumption habits however, a longitudinal in-situ evaluation measuring food waste as well is needed.

While we did see an increase in retention of items of which pictures were taken with the FoodFighters app, this was only based on a short-term memory test spanning two days. This timeframe is in hindsight short but for perishable food, even two days can make a significant difference. In future evaluations this should probably be increased to 5-7 days, and more importantly we need to evaluate whether it has an impact on timely consumption and the prevention of stockpiling.

By design FoodFighters encouraged extended interaction with food items. This, resulted in participants spending more time on the food items they converted into Food Fighters compared to the control group. It remains unclear if the higher retention was due to the extra time with the item or due to participants customizing and fight with their own characters, thereby making it more memorable as previous work predicted [48]. Another explanation could be that FoodFighters demands that the users engage with the food items in a tangible manner, which has been shown to help memory retention and learning as well [30, 32]. Future evaluations should investigate

these factors in more detail to quantify FoodFighters' actual impact on retention.

The biggest limitation this initial evaluation revealed was whether FoodFighters will be used regularly in the long-term. Despite all users appreciating the customization options, some theorized the long term enjoyment of the app might currently be limited. One particular element of the current iteration of FoodFighters that was criticised is the fighting stage, which is currently fully automated and deemed too long of a process by the participants. One option to overcome this, would be a manual mode that could for example follow the fighting system of Pokémon [28] where different attacks with different damage statistics are available at the varying levels of the fighters.

Participants also noted confusion about health and attack values for the Food Fighters. In our current version, these values were randomly assigned. However, for future versions this issue could be solved for example by detecting what food item [46] the user took a picture of and then base the values on the food's perishability (higher values to food that is more likely to spoil quickly). This would not only shift the users' focus to food items that go bad quickly and thereby making it more likely that they are used in time, but also might help establish some food literacy on perishability (if correctly communicated). Furthermore, in the current version users theoretically can also create food fighters out of non-food items, which would be thereby prevented as well. Similarly, for the customization options, in the next iterations FoodFighters could offer an increasing number of customization options over time, or customization elements that come as a reward for winning a battle or creating certain amounts of Food Fighters, to increase long-term interest. Furthermore, we can increase the potential long term use by adding more social features [50], for example allowing players to team up with friends battling against other players in group-battles.

One potential issue, that has not been addressed as part of the study is the way users will take pictures of their food items. For example, if they frequently open their fridge and leave it open for too long this might cause a significant waste of energy that need to be addressed. It could be that a small reminder in the picture taking stage would already be enough to help users remember to close the fridge quickly again, however for this, further studies in real-world settings are needed.

6 CONCLUSION

In this paper, we presented the serious game FoodFighters, a novel way to potentially reduce food waste through increasing memory retention of the users' food supplies. Users have to take a picture of a food item they have at home, then can customize it with options, such as arms, legs, eyes, etc. can transform it into a Food Fighter, and fight other players fighters. Here, the regular picture taking, is meant to help with memory retention. Our preliminary evaluation indicates - especially in combination with the customization step - that this indeed helped players recall 25% more food items if they had been made into Food Fighters. However, the current prototype has several limitations that need to be addressed. First and foremost the users wished for a more interactive fighting stage, as well as a clear reasoning behind the attack and health stat assigned to the food fighters. The next step is to implement these features

and then carry out a longitudinal deployment study over several weeks where not only long-time engagement but also the apps impact on food waste needs to be measured. Thereby, we hope that FoodFighters will be able to help with the reduction of food waste in a fun and engaging way.

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